

O-RAN Open Fronthaul S-Plane — Software Testbed

Configuration, Setup and Standards Reference · Revised 2026-09-17 · **validation figures superseded 2026-09-29** — see notice below

SUPERSEDED IN PART — revised 2026-09-29. The 2026-09-17 **132-run** figures below are **withdrawn**. A cross-check audit on 2026-09-20 found four defects in the campaign apparatus, two of which fabricated data — including that `B3_pdv_congestion` never injected anything (this kernel has no `sch_netem` and the failure was swallowed), which **invalidates B3 in the 132-run campaign as well**. All scenarios were re-executed as a **168-run** campaign (12 replicates × 14 scenarios), recounted from the raw per-run files on 2026-09-29.

Current authoritative figures (168 live runs, per-scenario macro averages): pre-frozen base rule — attack sensitivity **0.625**, specificity **0.783**, attribution **0.552**; post-observation `decision_rule_v3` (primary) — attack sensitivity **0.990**, specificity **0.783**, attribution **0.875**; abstention correctness **1.000**. Specificity is 0.783 because `B_bc_replacement` scores 0/12 (12 false ATTACK calls) and B3 abstains on 1 of 12; excluding `B_bc_replacement` it is **0.979**, not 1.000. `C1_removeal` is 11/12: the interception detector fires below a 0.500 delivered/declared rate ratio, caught 65 % blackhole (ratio 0.458–0.462) and missed 60 % (ratio 0.505).

Authority for these numbers: `splane_campaign_CORRECTED_2026-09-20.tgz` (sha256 6149b4fb...), `EVALUATION_V4.json`, and `ORAN_SPlane_AUDIT_AND_CORRECTIONS` (2026-09-29 revision). The configuration, topology, standards traceability, fault taxonomy and test-case definitions in this document are **unchanged and remain valid**; only the validation figures are superseded.

Document scope. Every hardware and software configuration used by the testbed, each traced to the standard or reference implementation that fixes it. No value is assumed — all were read back from the running system or from the cited standard.

Telecom profile. ITU-T G.8275.1 — PTP telecom profile for phase/time distribution with full timing support. Selected because O-RAN sync configs LLS-C1/C2/C3 distribute timing through the network, and because the real TIMESAFE hardware captures used for comparison run domain 24 (G.8275.1) on all 344,630 PTP packets across the six captures.

Frozen state. 18 artefacts SHA-256 hashed; `verify_frozen.sh` refuses to run if any changed and was passed before every admissible run.

Validation result as issued 2026-09-17 — NOW WITHDRAWN. 132 admissible runs across 12 randomised replicates. Attack sensitivity 1.000 [0.940, 1.000] (n=60); benign specificity 0.800 [0.682, 0.882] (n=60) — 1.000 [0.926, 1.000] (n=48) once the single known schema limitation `B_bc_replacement` is excluded, which accounts for all 12 false positives; honest abstention 12/12; attack attribution 0.817 [0.701, 0.894]. Zero infrastructure failures.

Corrections in this revision (all pre-freeze, all verified). (1) Added `dataset_comparison G.8275.x` — it was missing, so `linuxptp` had been running the IEEE-1588 *default* BMCA rather than the G.8275.1 *alternate* BMCA. A capture cannot reveal this (the BMCA algorithm is not a wire field), so the earlier "verified by parsing its own capture" could not have caught it. (2) Removed `path_trace_enabled 1` — real TIMESAFE captures carry zero TLVs in all 344,630 PTP packets of the six captures, so leaving `PATH_TRACE` on makes its presence a perfect testbed-vs-TIMESAFE separator (a provenance leak). (3) Added a per-node config generator (`run/mkcfg.sh`); option scoping verified against `linuxptp v4.0 config.c`. (4) A 4th provenance leak was logged: `minor_version_ptp` (testbed 4.0 = 1, real = 0) — a compile-time constant, apparatus-only, excluded from any join.

1 Hardware / platform (software testbed)

Item	Value (read from the system)	How verified	Consequence
Kernel	Firecracker microVM	<code>uname -r</code>	Monolithic build
Privileges	uid 0, CAP_NET_ADMIN	<code>id -u;</code> <code>capsh --print</code>	Namespaces and bridges permitted
Loadable modules	None (no /lib/modules)	<code>modprobe</code> <code>ptp_mock</code> → not found	No per-node mock PHC
PTP hardware clock	None (no /dev/ptp*)	<code>ls</code> <code>/dev/ptp*</code>	No hardware clock to discipline
veth timestamping	software only; PHC: none	<code>ethtool -T</code>	Microsecond noise floor vs a nanosecond target (measured: master offset median 0.9 μ s, p95 5.6 μ s; path delay median 7.2 μ s, p95 8.3 μ s (792 samples from the 132-run campaign's baseline runs))
Oscillator	One shared kernel clock	Architectural	Clocks cannot drift apart; B1/B6 impossible

A second physical machine would add a second crystal (makes oscillator drift B6 real); a hardware-timestamping NIC + back-to-back cable drops the floor from μ s to tens of ns (makes A4/B5 measurable). Neither unlocks GNSS (A6/A7/B1) or SyncE (B4).

2 G.8275.1 profile configuration (Class A — exact)

Contents of `cfg/g82751.base`, applied to every node. Cross-checked line-by-line against linuxptp v4.0's own `configs/G.8275.1.cfg` and against the real TIMESAFE capture.

linuxptp key	Value	Standard / basis	Observed in real capture
dataset_comparison	G.8275.x	G.8275.1 alternate BMCA (ADDED this revision)	algorithm, not a wire field
domainNumber	24	G.8275.1 — telecom range	24 on all

		24–43	344,630 PTP pkts ✓
logSyncInterval	-4 (16/s)	G.8275.1 mandated Sync rate	-4 on all Sync ✓
logAnnounceInterval	-3 (8/s)	G.8275.1 mandated Announce rate	-3 on all Announce ✓
logMinDelayReqInterval	-4 (16/s)	G.8275.1	—
announceReceiptTimeout	3	G.8275.1 (375 ms at 8/s)	—
priority1	128	G.8275.1 fixes it and removes it from BMCA compare	128 on legitimate clocks ✓
delay_mechanism	E2E	G.8275.1 requires end-to-end	Delay_Req/ Delay_Resp present ✓
network_transport	L2	G.8275.1 — Ethernet, no IP	EtherType 0x88F7 ✓
ptp_dst_mac	01:1B:19 :00:00:00	G.8275.1 forwardable multicast	100% conformant ✓
path_trace_enabled	1	REMOVED this revision — PATH_TRACE provenance leak (real capture has 0 TLVs)	absent in real capture
G.8275.*.localPriority	128	G.8275.1 alternate-BMCA tiebreaker	—
maxStepsRemoved	255	IEEE 1588 / G.8275.1	—

Why this matters most. The project's earlier netem runs used domain 0, logSync -3, logAnnounce +1 (one Announce per 2 s vs the standard's 125 ms — a 16× mismatch). Because benign data came from that testbed and attack data from real G.8275.1 hardware, a classifier could have separated the two on cadence alone. Aligning the testbed to G.8275.1 and adding the alternate-BMCA comparison removes that confound.

3 Declared deviations from a production deployment

Setting	Why it differs	What it costs
<code>free_running 1</code>	All namespaces share one kernel clock; with the servo active they would fight over the same clock.	Offsets are measured and logged but not applied. Servo convergence dynamics absent.
<code>time_stamping software</code>	veth exposes no PHC (<code>ethtool -T</code>).	Microsecond floor. No G.8273.2 / G.8271.1 conformance claim.
<code>clock_classes_threshold 248</code>	An acceptance policy, not a quality claim; the lab has no PRTC/GPS so every node truthfully advertises clockClass 248.	No node given a fabricated clockClass 6; hierarchy uses priority2 (an operator field by design).

4 Topology and node roles

Two Ethernet bridges separated by a Boundary Clock (models O-RAN LLS-C2/C3).

```
[gma p2=100] [gmb p2=110] --- brUP --- [bc: v-bc-up | v-bc-dn] --- brDN --- [ru1] [ru2] [ru3]
```

Node	Role	priority2	Purpose
gma	Primary grandmaster (T-GM)	100 (best)	Wins BMCA normally
gmb	Provisioned backup GM	110	Enables B2 planned failover
bc	Boundary clock (T-BC)	120	stepsRemoved 0 → 1; enables A8 and B7
ru1/ ru2/ ru3	O-RU proxies (T-TSC)	128	Three clients = spatial axis

5 Scenario configurations (11 — randomised per replicate)

ID	Scenario	How produced on the wire	Class
—	baseline	All nodes nominal, GM-A → BC → 3 RUs	healt

hy

A1	Rogue grandmaster	ptp4l instance, randomised off-allow-list identity, randomised superior priority2 (1–90) and clockClass (6/7/13), self-announcing on RU segment	attack
A2	Sync spoof	scapy forged Sync (two-step flag set), randomised implausible originTimestamp and burst, unprovisioned source; never Announce	attack
A3	Replay	Live slice captured, retransmitted with tcpreplay (stale timestamps, reused sequenceId); randomised slice size and loop count	attack
A5	DoS / flood	scapy high-rate PTP flood (randomised burst/rate)	attack
A8	Rogue boundary clock	Two-port ptp4l (randomised priority2 105–118) slaving upstream, relaying downstream with stepsRemoved incremented	attack
B2	Planned GM failover	GM-A withdrawn, GM-B (on allow-list) elected, maintenance window open	benign

B_UNPLANNED	Unplanned GM crash (NEW)	GM-A killed, NO maintenance window; GM-B elected. Packet-indistinguishable from a suppression attack → expected verdict UNKNOWN (honest abstention)	benign
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B_BCREPL	BC replacement (NEW)	In-path BC swapped for a provisioned replacement during maintenance. The replacement id is provisioned (expected_bc_identity_secondary), but the frozen rule reads only one BC identity → flagged A8; reported, not patched	benign
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B3	PDV / congestion	tbf bottleneck on the BC → RU timing path + competing non-PTP background traffic, randomised (sch_netem is unavailable in this kernel)	benign
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B7	Topology change	RU3 link bounced and re-homed during maintenance	benign
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A1 vs B2 is the decisive pair. Both present a new grandmaster identity; they are separated only by provisioned context (allow-list + maintenance window). Without that context they are indistinguishable from the packets alone.

6 Faults deliberately out of scope (physical reason)

Fault	Why not producible/measurable here	What would be required
A6 GNSS spoof / A7 jam	No GNSS receiver in software; faking a status enum is inventing evidence	GNSS simulator + receiver + RF-shielded chamber
B1 GNSS holdover	No reference to lose, no characterised oscillator	GNSS timing receiver + independent oscillator
B4 SyncE / EEC	Physical-layer function; veth has no PHY	SyncE-capable NIC + switch, sync4l
B6 Oscillator drift	All namespaces share one crystal	A second physical machine
A4 Delay attack	Injectable, but a passive capture yields only t1,t4 (t2,t3 never on the wire)	Live per-client pmc telemetry or a redundant path; HW timestamping for ns claims

7 Validity controls

Control	How enforced
No tunable parameters	Every constant in the decision rule is a standard, a provisioned fact, or a structural definition. The one self-chosen definition is the 1% transient filter (<code>TRANSIENT_FRACTION = 0.01</code>). Its standard replacement (IEEE 1588 §9.3.2.4, <code>FOREIGN_MASTER_THRESHOLD = 2</code> within 4 Announce intervals) has been identified but not adopted, because adopting it needs a re-freeze
Pre-registration	<code>freeze.sh</code> SHA-256 hashes 18 artefacts; <code>verify_frozen.sh</code> passed before every admissible replicate
Development /validation split	Reps that exposed defects are development data and are excluded; only post-freeze reps (30–41) are scored
Randomisation	Per-rep attacker identities, priorities, timings and burst sizes (<code>run/randparams.py</code> , seeded by rep) — runs are genuinely independent, so CIs are honest not optimistic
Session-disjoint splitting	Evaluation by run, never by window
Fail-closed	Missing/unreadable evidence → UNKNOWN, never BENIGN; infrastructure

Extension to the scenario tables (pp. 5-6): source, destination, attack devices, consequence

Added 5 Oct 2026. This page extends the table on the preceding page with four columns requested after the PRISM review: **Source** (who or what originates the attack or fault), **Destination** (where it hits in the O-RAN architecture), **Attack devices** (devices that could originate it; for benign faults, the originating devices) and **Consequence**. All pages before and after this one are unchanged from the 29 Sep 2026 revision.

The 14 executed scenarios

ID	Fault / scenario	Source (of attack or fault)	Destination (where it hits in the architecture)	Attack devices (possible originating devices)	Consequence (effect)
A1	A1 rogue master	Internal attacker on the fronthaul timing network posing as a grandmaster: forged/superior Announce data set wins the BMCA [R7384 s3.2.4 rogue master; TR106 T-SPLANE-02/03; TS105 11.1.5.2.2].	O-RU PTP slave port - its BMCA / parent selection on the Open Fronthaul S-plane; every O-RU on the attacker's segment re-parents to the rogue [TS982; MEAS: 36/36 RU logs selected the rogue as best master].	Any device that can send PTP on the fronthaul segment: compromised O-DU (LLS-C1/C2), compromised fronthaul switch or boundary clock, compromised O-RU, or an unauthorised host plugged into the fronthaul Ethernet at the cell site; supply-chain backdoored DU/RU [TSAFE s4; TR106 T-FRHAUL-02]. Testbed: second ptp4l in namespace 'rogue' on brDN.	False time distributed to all downstream O-RUs [R7384 Table 1]. Published impact in one experiment: O-RU crash ~2 s after attack start with switch PTP ports in dynamic role, 5G cell drop, manual reboot [TSAFE]. At stake: cell phase sync < 3 us, TAE 65 ns MIMO [3GPP].
A2	A2 sync spoof	Attacker injecting forged Sync/Follow_Up (false originTimestamp / correctionField) under an off-allow-list identity [R7384 s3.2.2 spoofing].	O-RU PTP slave port - the timestamp content its servo consumes [TS982].	Any host able to transmit on the fronthaul segment: compromised O-RU, O-DU, switch, or an attached rogue host [TSAFE s4; TR106 T-FRHAUL-02]. Testbed: scapy inject.py from the ru3 namespace.	False time / accuracy degradation if accepted [R7384 Table 1]. Measured: no master change in any RU log; visible only in packet evidence [MEAS].
A3	A3 replay	Man-in-the-middle or injector recording and re-sending genuine PTP frames later (stale timestamps, reused sequenceId) [R7384 s3.2.3 replay].	O-RU PTP slave port - sequence continuity and timestamp freshness of the master's messages [TS982].	Device with capture + transmit access on the segment: compromised switch/port mirror, compromised O-RU/O-DU, attached host [TSAFE s4.1 replay]. Testbed: tcpdump + tcp replay from ru3.	False time / accuracy degradation if stale messages are accepted [R7384 Table 1; TSAFE replay]. Measured: no master change in any RU log; visible as sequenceId regressions [MEAS].
A5	A5 DoS flood	Flood of PTP traffic at the master/slave or segment [R7384 s3.2.9 DoS against the time protocol; TR106 T-SPLANE-01; TS105 11.1.5.1.1, 24.2.1.1].	PTP processing of the master clock and of the O-RU ports on the segment [TR106 T-SPLANE-01; TS982].	Any device able to transmit on the segment, including a compromised O-RU or an attached host [TSAFE s4]. Testbed: flood.py from ru3 (~38,000 records/run vs ~6,000 baseline).	Clock service interrupted / degraded [TR106 T-SPLANE-01; R7384 Table 1 DoS]. Measured: no master change in any RU log at the rates used [MEAS].
A8	A8 rogue BC	Rogue boundary clock inserted in the path, or an authorised BC/TC turned malicious: relays the real GM downstream under its control (campaign: stepsRemoved +1) [R7384 s3.2.4 in MITM position; TR106 T-SPLANE-03].	O-RU parent selection - every O-RU behind the inserted clock [TR106 T-SPLANE-03; MEAS: 36/36 RU logs took the rogue BC as new master].	Compromised boundary clock or fronthaul switch with PTP, or a two-port host bridging both segments [TSAFE s4; TR106 T-FRHAUL-02]. Testbed: two-port ptp4l in namespace 'rbc'.	Attacker controls the time relayed to all downstream O-RUs; slaves given inaccurate time [TR106 T-SPLANE-03; R7384 Table 1 false time].
C1	C1 packet removal	Man-in-the-middle selectively intercepting and dropping PTP packets [R7384 s3.2.5; TR106 T-SPLANE-04; TS105 11.1.5.3.1].	Downstream port of the boundary clock -> the O-RUs it serves (delivery of Sync/Announce) [TS982].	In-path element: compromised fronthaul switch or boundary clock; external MITM on the link also possible [R7384 s3.1, Table 1]. Testbed: BC downstream port blackholed.	Accuracy loss or free-running / DoS [TR106 T-SPLANE-04]. Measured: 36/36 RU ports lost master on Announce timeout; servo reports 7.8 vs 22 per log [MEAS].
C2	C2 malformed	Injection of structurally invalid PTP frames (fuzzing) to exploit parser weaknesses [R7384 s3.2.11; TS105 24.2.1.2].	PTP parser / stack of the O-RU and other clocks on the segment [TS105 24.2.1.2].	Any host able to transmit on the segment, including a compromised O-RU [TSAFE s4]. Testbed: inject_malformed.py from ru3.	Implementation-dependent: from silent discard to stack crash / DoS [R7384 s3.2.11]. Measured: no master change in any RU log [MEAS].
C3	C3 whole-second	Forged Announce abusing time-properties fields: leap61 set, currentUtcOffset 0 instead of 37, traceability stripped [R7384 s3.2.2 spoofing].	O-RU time-properties data set (UTC offset / leap handling) [TS982].	Any host able to transmit on the segment under the GM's identity [TSAFE s4]. Testbed: inject_wholesecond.py from ru3.	False time by whole seconds in UTC-derived timing [R7384 Table 1 false time]. Measured: no master change; servo reports 3.0 vs 22 per log [MEAS].
baseline	baseline	None (healthy control).	None.	None.	None - reference behaviour.
B2	B2 GM failover	Operator maintenance: primary grandmaster withdrawn during an open maintenance window - no attacker.	BMCA at every clock; O-RUs re-parent to the allow-listed backup GM [G8275].	Originating devices (benign): GM-A (withdrawn), GM-B (provisioned backup).	Brief re-parenting; time stays traceable to an allow-listed GM. Packet-identical to A1 except for allow-list and maintenance context [MEAS: 12/12 BENIGN].
B3	B3 congestion	Competing data-plane traffic queuing PTP behind it in the fronthaul (congestion) - no attacker [G8261].	PTP packets on the BC -> O-RU path (queueing delay at the bottleneck).	Originating device (benign): congested fronthaul switch/link queue. Testbed: tbf 1 Mbit bottleneck on v-bc-dn plus background traffic.	Packet delay variation rises (measured Sync PDV 0.07 ms -> 5.6-857 ms), offset noise grows; 12/36 RU logs hit the Announce timeout - so a timeout alone does not prove an attack [MEAS; G8261].
B7	B7 topology	Planned topology change: a link bounced / O-RU re-homed during maintenance - no attacker.	The affected O-RU's port state and path (stepsRemoved, path delay) [G8275].	Originating devices (benign): fronthaul link / switch port being reconfigured.	The affected O-RU re-parents after the link returns; stepsRemoved and path delay change consistently with the new topology and no field becomes illegal [MEAS: 12/12 BENIGN].

ID	Fault / scenario	Source (of attack or fault)	Destination (where it hits in the architecture)	Attack devices (possible originating devices)	Consequence (effect)
B_bc_replacement	B_bc_replacement	Planned replacement of the boundary clock during maintenance; replacement identity provisioned - no attacker.	O-RU parent selection (new BC identity in the path).	Originating devices (benign): old BC (removed), provisioned replacement BC.	O-RUs lose the old BC (36/36 RU logs hit the Announce timeout during the swap) and re-parent to the replacement [MEAS]. Structurally identical to A8, which is why the base rule mis-reads it (0/12).
B_unplanned_failover	B_unplanned_failover	Grandmaster crash with no maintenance window open - cause not knowable from packets (failure or attack).	BMCA at every clock; O-RUs re-parent to the provisioned backup GM.	Originating devices: GM-A (failed), GM-B (provisioned backup). Intent unknown.	Re-parenting to an allow-listed GM; time stays traceable. Correct handling is escalation (UNKNOWN), not a guess [MEAS: 12/12 UNKNOWN].

Classes not producible on the software testbed

ID	Fault / scenario	Source (of attack or fault)	Destination (where it hits in the architecture)	Attack devices (possible originating devices)	Consequence (effect)
A4	A4 delay attack	In-path attacker delaying PTP messages in one direction only (engineered asymmetry) [R7384 s3.2.6 delay; TS105 11.1.5.3.2].	O-RU offset computation - PTP assumes symmetric paths [1588].	Any in-path forwarding element: fronthaul switch, boundary clock, transparent clock, compromised router [R7384 s3.1 MITM].	Constant time error equal to half the injected asymmetry, invisible to field checks [1588; R7384 Table 1 accuracy degradation]. Not run here: needs netem / hardware timestamps.
A6	A6 GNSS spoof	Coherent GNSS spoofing at the grandmaster's antenna [R7384 s3.2.10 grandmaster time-source attack; BSI].	PRTC/T-GM GNSS receiver - the root time reference of the whole S-plane [R7384 s3.2.10].	RF transmitter / GNSS simulator near the GM or (LLS-C4) O-RU antenna [BSI; R7384 s3.2.10].	Whole-network false time [R7384 s3.2.10]; every PTP field stays legal, so packet checks cannot see it [MTC; deck slide 37]. Hardware-only; not produced here.
A7	A7 GNSS jam	GNSS jamming at the grandmaster's antenna [R7384 s3.2.10; BSI].	PRTC/T-GM GNSS receiver [R7384 s3.2.10].	RF jammer near the GM or (LLS-C4) O-RU antenna [BSI].	GM loses its reference and enters holdover (clockClass 7, then out-of-holdover-spec classes) [G8275]; time error grows at the holdover oscillator's rate. Hardware-only; not produced here.
B1	B1 GNSS holdover	Honest loss of GNSS at the grandmaster (sky obstruction, antenna/cable fault, weather) - no attacker.	T-GM time reference; announced downstream as clockClass 6 -> 7 [G8275].	Originating device (benign): T-GM GNSS receiver / antenna.	Orderly, declared degradation: time error grows at the holdover oscillator's rate; clockClass moves to out-of-holdover-spec classes if the budget is exceeded [G8275; G8271]. Hardware-only here.
B4	B4 SyncE / EEC	Degradation of the physical-layer frequency reference (SyncE / EEC, ESMC quality level) - no attacker.	Frequency synchronisation of each node's equipment clock.	Originating devices (benign): SyncE-capable PHYs / EECs along the path.	Frequency wander; phase drifts faster when PTP is lost (weaker hybrid holdover). Hardware-only (veth has no PHY).
B5	B5 static asymmetry	Static path asymmetry (unequal fibre lengths, asymmetric equipment latency) - no attacker.	O-RU offset computation (symmetric-path assumption) [1588].	Originating devices (benign): fibre/transport links, transponders.	Constant time error equal to half the asymmetry [1588]; not visible in any PTP field. Detection logic only in software.
B6	B6 oscillator drift	Free-running local oscillator frequency error (temperature, ageing) - no attacker.	Each clock's local oscillator; matters most in holdover / free-run.	Originating devices (benign): crystal/TCXO/OCXO of the O-RU, BC or GM.	Time error accumulates at the frequency offset. Measured between two consumer laptop crystals: +21.013 ppm relative [20.623, 21.403], i.e. ~21 us of separation per second, which would exhaust a 1.5 us budget in ~0.07 s without discipline [B6M; G8271]. Telecom-class oscillators are far better; this is not a telecom measurement.
B8	B8 SW timestamp noise	Software timestamping noise from OS scheduling - a measurement artefact, no attacker.	The time-error measurement itself (all clocks in a software testbed).	Originating device (benign): host kernel / NIC without a PTP hardware clock.	Microsecond-scale noise floor against a nanosecond target; limits any absolute time-error claim.

Key to bracketed tags: **R7384** IETF RFC 7384 (2014): s3.1 attacker classes, s3.2 threats, Table 1 impact (false time / accuracy degradation / DoS) · **TR106** ETSI TR 104 106 V3.0.0 (2025-06) O-RAN threat model: s7.2 threat agents; T-SPLANE-01...04; T-FRHAUL-01/02 · **TS105** ETSI TS 104 105 V7.0.0 (2025-06) O-RAN security test specification (test-case IDs) · **TS982** ETSI TS 103 982 V8.0.0 O-RAN architecture: O-RU hosts Low-PHY + RF and terminates Open Fronthaul (cl. 3.1, 6.3.6) · **TSafe** Groen et al., TIMESAFE, ACM TOPS 28(5) 2025 / arXiv 2412.13049v3: s4 access paths and attacks; O-RU crash ~2 s after attack start with switch PTP ports in dynamic role (one observed experiment, not a standardised limit) · **G8275** ITU-T G.8275.1 telecom profile clockClass semantics via Calnex one-page summary: 6 T-GM locked, 7 T-GM holdover in spec, 135 T-BC holdover in spec, 140/150/160 T-GM out of holdover spec, 165 T-BC out of holdover spec, 248 no reference since start-up · **G8271** ITU-T G.8271.1 network time-error budget: 1.5 us at reference point E, 1100 ns network limit at point C (v8 deck slide 7; Calnex budget summary) · **G8261** ITU-T G.8261 packet delay variation (PDV) context · **3GPP** 3GPP TS 38.133 cell phase sync < 3 us (via Ruffini, ATIS 2018); TS 38.104 cl. 9.6.3.2 TAE 65 ns MIMO · **BSI** BSI 5G Risk Analysis: Radio Access Network, v1.0 (2025) - GNSS jamming/spoofing · **1588** IEEE 1588-2019 offset/delay computation (asymmetry error = half the path asymmetry) · **MEAS** Measured in this project (168-run campaign, RU1-RU3 ptp4l logs, 36 logs per scenario) · **B6M** Measured in this project: two-laptop crystal comparison, 2 Oct 2026, outputs/B6_two_machine_2026-10-02/run2/DRIFT_REPORT_run2.md (run 2 ESTABLISHED, +21.013 ppm [20.623, 21.403]; run 1 NOT ESTABLISHED, +20.161 ppm; the 0.852 ppm run-to-run difference exceeds either CI). Consumer laptop crystals against an NTP reference - not telecom-class oscillators · **MTC** this catalogue's own entry for the family.

failures retried once then discarded with disclosure

Leak audit `run/leak_check.py` checks every field for testbed-vs-real separability; 4 leaks found: 3 eliminated by the G.8275.1 testbed (16× cadence confound, synthetic offsetScaledLogVariance 4096-vs-65535, PATH_TRACE), and 1 that cannot be fixed by configuration (minor_version_ptp) excluded from any testbed+real join

8 Validation result — SUPERSEDED (see notice on page 1)

Metric	Value	95% Wilson CI	n
Sensitivity (attack recall)	1.000	[0.940, 1.000]	60
Specificity — all benign runs (TN 48, FP 12)	0.800	[0.682, 0.882]	60
Specificity — excluding the known B_bc_replacement schema limitation	1.000	[0.926, 1.000]	48
Abstention correctness (ambiguous benign → UNKNOWN)	1.000	[0.757, 1.000]	12
Attack attribution (correct fault id)	0.817	[0.701, 0.894]	60

Binary attack detection 60/60 (TP=60, FN=0). All 12 specificity false positives are the single B_bc_replacement schema limitation. Attribution confusion A1 → A3 (×4) and A8 → A3 (×7) is the documented re-parenting effect (rogue master/BC insertion causes real sequenceId restarts, caught by the A3 clause first); it does not change the binary ATTACK verdict. Cross-domain: the frozen rule, applied unchanged to six real TIMESAFE hardware captures it never saw, classified **6 of 6 as ATTACK**; **5 of 6** on standards/protocol rules alone (3 by Announce-field legality, 2 by IEEE 1588 cl.11.3 sequenceId monotonicity). The remaining one is a healthy-looking spoof with fully conformant fields and needs the provisioned allow-list.

Artefacts: results_2026-09-17/ (EVALUATION.json, TIMESAFE_CROSSVAL.json, FROZEN.json, SESSION_FINDINGS.md, g87251_testbed_v2.tgz, full 132-run raw archive). Prior 81-run revision retained in _pre_2026-09-17_doc_update/.